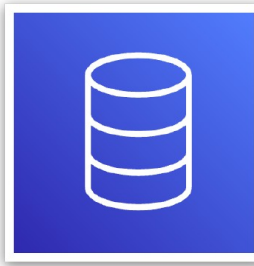


- Traditional applications leverage RDBMS databases
- These databases have the SQL query language
- Strong requirements about how the data should be modeled
- Ability to do query joins, aggregations, complex computations
- Vertical scaling (getting a more powerful CPU / RAM / IO)
- Horizontal scaling (increasing reading capability by adding EC2 / RDS Read Replicas)

Database Types



- **RDBMS (= SQL / OLTP):** RDS, Aurora – great for joins
- **NoSQL database – no joins, no SQL :** DynamoDB (~JSON), ElastiCache (key / value pairs), Neptune (graphs), DocumentDB (for MongoDB), Keyspaces (for Apache Cassandra)
- **Object Store:** S3 (for big objects) / Glacier (for backups / archives)
- **Data Warehouse (= SQL Analytics / BI):** Redshift (OLAP), Athena, EMR
- **Search:** OpenSearch (JSON) – free text, unstructured searches
- **Graphs:** Amazon Neptune – displays relationships between data
- **Ledger:** Amazon Quantum Ledger Database
- **Time series:** Amazon Timestream

- Note: some databases are being discussed in the Data & Analytics section

Amazon RDS – Summary



- Managed PostgreSQL / MySQL / Oracle / SQL Server / DB2 / MariaDB / Custom
- Provisioned RDS Instance Size and EBS Volume Type & Size
- Auto-scaling capability for Storage
- Support for Read Replicas and Multi AZ
- Security through IAM, Security Groups, KMS , SSL in transit
- Automated Backup with Point in time restore feature (up to 35 days)
- Manual DB Snapshot for longer-term recovery
- Managed and Scheduled maintenance (with downtime)
- Support for IAM Authentication, integration with Secrets Manager
- RDS Custom for access to and customize the underlying instance (Oracle & SQL Server)

- **Use case:** Store relational datasets (RDBMS / OLTP), perform SQL queries, transactions

Amazon Aurora – Summary



- Compatible API for PostgreSQL / MySQL, separation of storage and compute
- Storage: data is stored in 6 replicas, across 3 AZ – highly available, self-healing, auto-scaling
- Compute: Cluster of DB Instance across multiple AZ, auto-scaling of Read Replicas
- Cluster: Custom endpoints for writer and reader DB instances
- Same security / monitoring / maintenance features as RDS
- Know the backup & restore options for Aurora
- **Aurora Serverless** – for unpredictable / intermittent workloads, no capacity planning
- **Aurora Global**: up to 16 DB Read Instances in each region, < 1 second storage replication
- **Aurora Machine Learning**: perform ML using SageMaker & Comprehend on Aurora
- **Aurora Database Cloning**: new cluster from existing one, faster than restoring a snapshot
- **Use case**: same as RDS, but with less maintenance / more flexibility / more performance / more features

Amazon ElastiCache – Summary



- Managed Redis / Memcached (similar offering as RDS, but for caches)
- In-memory data store, sub-millisecond latency
- Select an ElastiCache instance type (e.g., cache.m6g.large)
- Support for Clustering (Redis) and Multi AZ, Read Replicas (sharding)
- Security through IAM, Security Groups, KMS, Redis Auth
- Backup / Snapshot / Point in time restore feature
- Managed and Scheduled maintenance
- Requires some application code changes to be leveraged
- **Use Case:** Key/Value store, Frequent reads, less writes, cache results for DB queries, store session data for websites, cannot use SQL.

Amazon DynamoDB – Summary



- AWS proprietary technology, managed serverless NoSQL database, millisecond latency
- Capacity modes: provisioned capacity with optional auto-scaling or on-demand capacity
- Can replace ElastiCache as a key/value store (storing session data for example, using TTL feature)
- Highly Available, Multi AZ by default, Read and Writes are decoupled, transaction capability
- DAX cluster for read cache, microsecond read latency
- Security, authentication and authorization is done through IAM
- Event Processing: DynamoDB Streams to integrate with AWS Lambda, or Kinesis Data Streams
- Global Table feature: active-active setup
- Automated backups up to 35 days with PITR (restore to new table), or on-demand backups
- Export to S3 without using RCU within the PITR window, import from S3 without using WCU
- Great to rapidly evolve schemas
- **Use Case:** Serverless applications development (small documents 100s KB), distributed serverless cache

Amazon DynamoDB



- Fully managed, highly available with replication across multiple AZs
- NoSQL database - not a relational database
- Scales to massive workloads, distributed database
- Millions of requests per seconds, trillions of row, 100s of TB of storage
- Fast and consistent in performance (low latency on retrieval)
- Integrated with IAM for security, authorization and administration
- Enables event driven programming with DynamoDB Streams
- Low cost and auto-scaling capabilities
- Standard & Infrequent Access (IA) Table Class

DynamoDB - Basics

Amazon DynamoDB

NoSQL Serverless Database

- DynamoDB is made of **Tables**
- Each table has a **Primary Key** (must be decided at creation time)
- Each table can have an infinite number of items (= rows)
- Each item has **attributes** (can be added over time – can be null)
- Maximum size of an item is **400KB**
- Data types supported are:
 - **Scalar Types** – String, Number, Binary, Boolean, Null
 - **Document Types** – List, Map
 - **Set Types** – String Set, Number Set, Binary Set

DynamoDB in Big Data

- Common use cases include:
 - Mobile apps
 - Gaming
 - Digital ad serving
 - Live voting
 - Audience interaction for live events
 - Sensor networks
 - Log ingestion
 - Access control for web-based content
 - Metadata storage for Amazon S3 objects
 - E-commerce shopping carts
 - Web session management
- Anti Pattern
 - Prewritten application tied to a traditional relational database: use RDS instead
 - Joins or complex transactions
 - Binary Large Object (BLOB) data: store data in S3 & metadata in DynamoDB
 - Large data with low I/O rate: use S3 instead

Amazon S3 – Summary



- S3 is a... key / value store for objects
- Great for bigger objects, not so great for many small objects
- Serverless, scales infinitely, max object size is 5 TB, versioning capability
- **Tiers:** S3 Standard, S3 Infrequent Access, S3 Intelligent, S3 Glacier + lifecycle policy
- **Features:** Versioning, Encryption, Replication, MFA-Delete, Access Logs...
- **Security:** IAM, Bucket Policies, ACL, Access Points, Object Lambda, CORS, Object/Vault Loc
- **Encryption:** SSE-S3, SSE-KMS, SSE-C, client-side, TLS in transit, default encryption
- **Batch operations** on objects using S3 Batch, listing files using S3 Inventory
- **Performance:** Multi-part upload, S3 Transfer Acceleration, S3 Select
- **Automation:** S3 Event Notifications (SNS, SQS, Lambda, EventBridge)
- **Use Cases:** static files, key value store for big files, website hosting

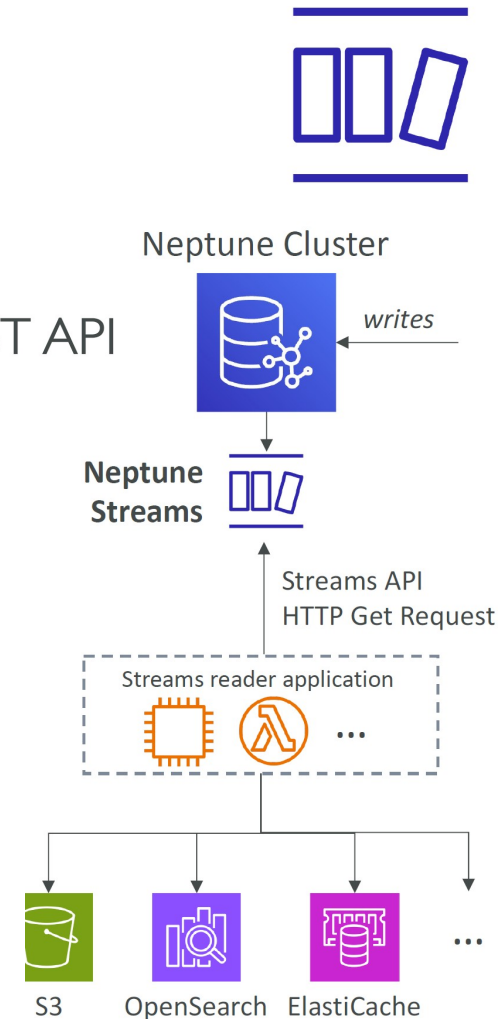
DocumentDB



- Aurora is an “AWS-implementation” of PostgreSQL / MySQL ...
- DocumentDB is the same for MongoDB (which is a NoSQL database)
- MongoDB is used to store, query, and index JSON data
- Similar “deployment concepts” as Aurora
- Fully Managed, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB
- Automatically scales to workloads with millions of requests per seconds

Amazon Neptune

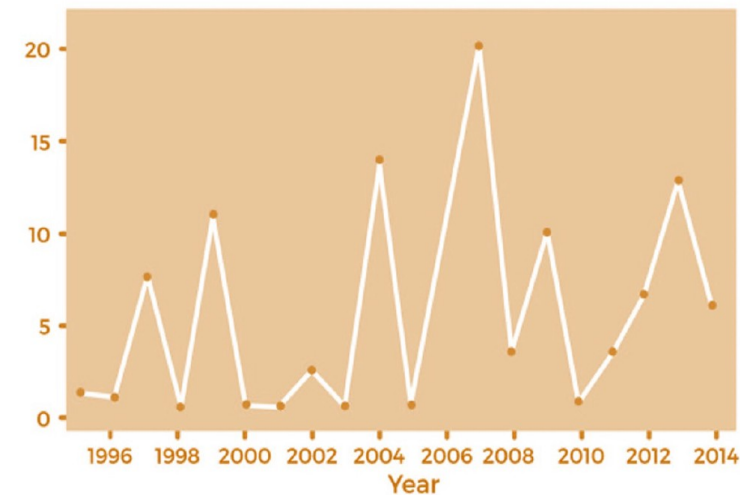
- Fully managed **graph** database
- A popular **graph dataset** would be a **social network**
 - Users have friends No duplicates, strict order
 - Posts have comments Streams data is accessible in an HTTP REST API
 - Comments have likes from users
 - Users share and like posts...
- Highly available across 3 AZ, with up to 15 read replicas
- Build and run applications working with highly connected datasets – optimized for these complex and hard queries
- Can store up to billions of relations and query the graph with milliseconds latency
- Highly available with replications across multiple AZs
- Great for knowledge graphs (Wikipedia), fraud detection, recommendation engines, social networking



Amazon Timestream



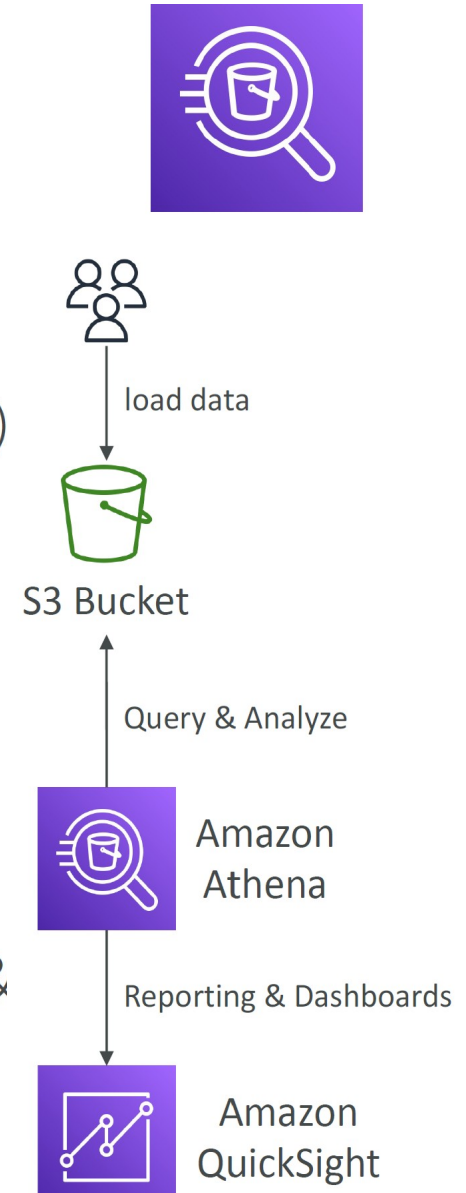
- Fully managed, fast, scalable, serverless **time series database**
- Automatically scales up/down to adjust capacity
- Store and analyze trillions of events per day
- 1000s times faster & 1/10th the cost of relational databases
- Scheduled queries, multi-measure records, SQL compatibility
- Data storage tiering: recent data kept in memory and historical data kept in a cost-optimized storage
- Built-in time series analytics functions (helps you identify patterns in your data in near real-time)
- Encryption in transit and at rest
- Use cases: IoT apps, operational applications, real-time analytics, ...



Data & Analytics

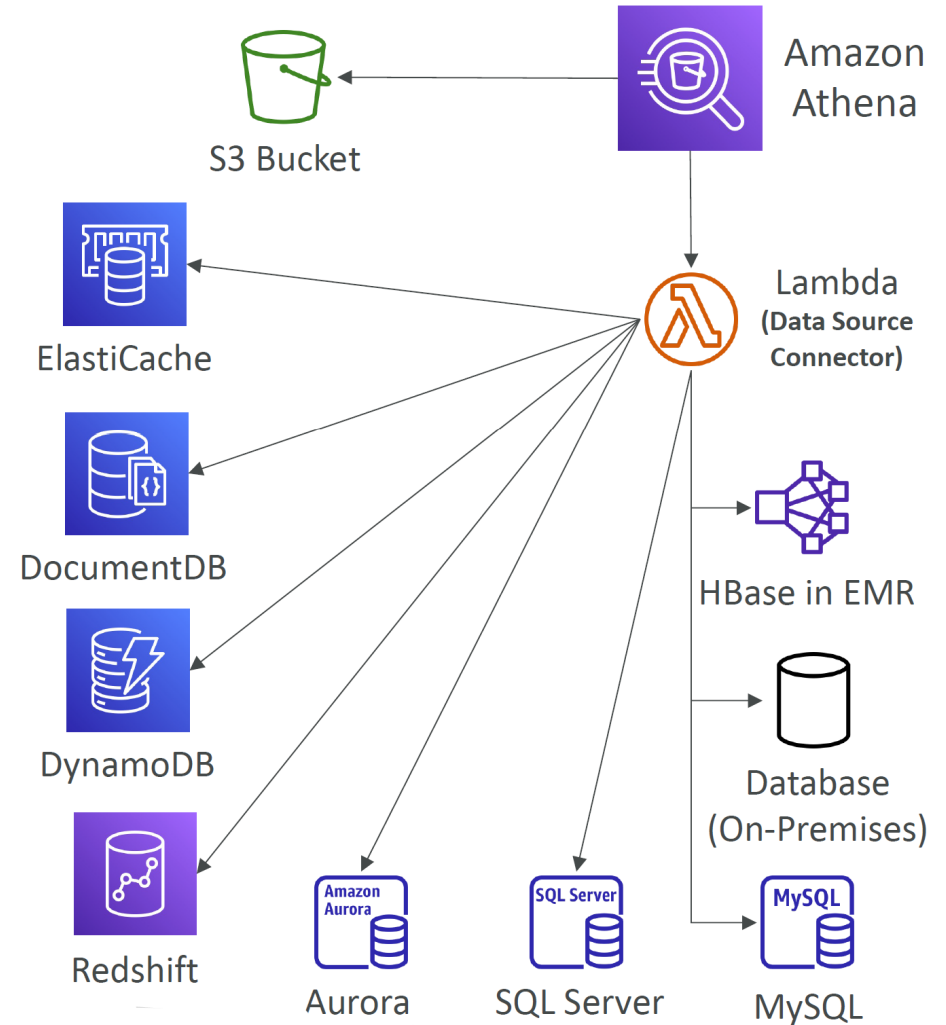
Amazon Athena

- **Serverless** query service to analyze data stored in Amazon S3
- Uses standard SQL language to query the files (built on Presto)
- Supports CSV, JSON, ORC, Avro, and Parquet
- Pricing: \$5.00 per TB of data scanned
- Commonly used with Amazon Quicksight for reporting/dashboards
- **Use cases:** Business intelligence / analytics / reporting, analyze & query VPC Flow Logs, ELB Logs, **CloudTrail trails**, etc...
- **Exam Tip:** analyze data in S3 using serverless SQL, use Athena



Amazon Athena – Federated Query

- Allows you to run SQL queries across data stored in relational, non-relational, object, and custom data sources (AWS or on-premises)
- Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g., CloudWatch Logs, DynamoDB, RDS, ...)
- Store the results back in Amazon S3



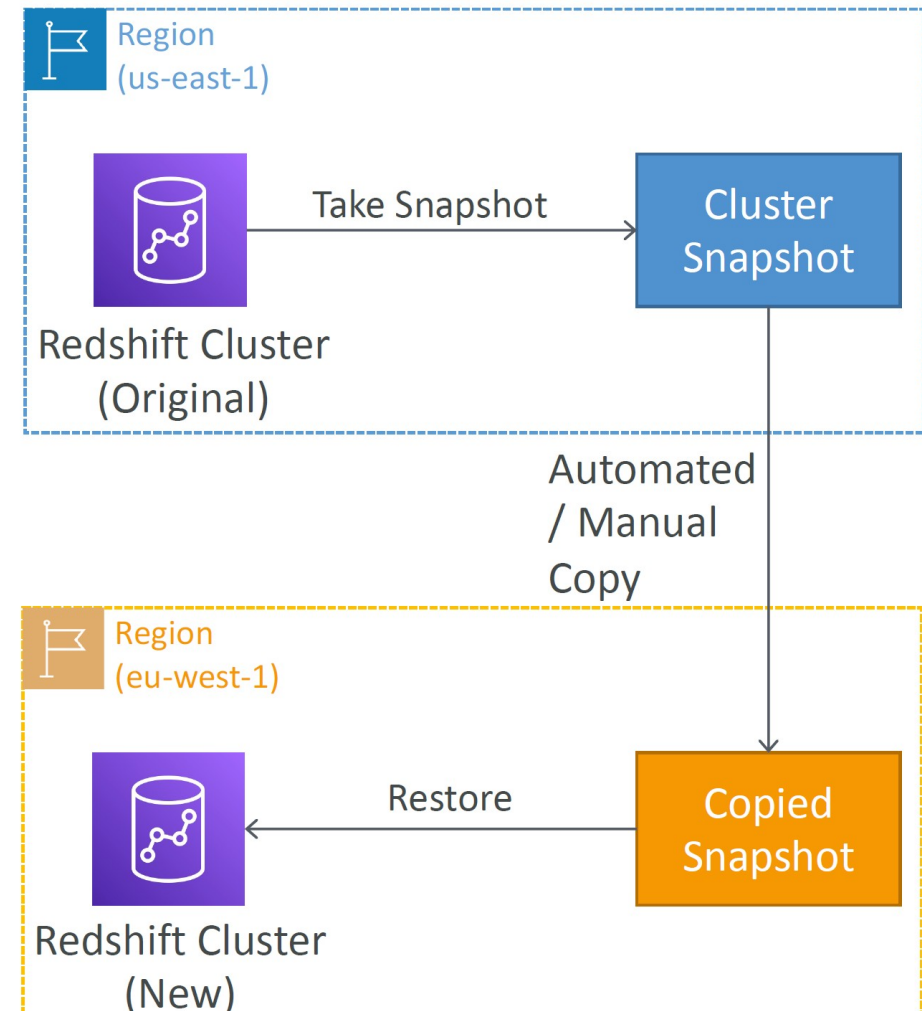
Redshift Overview



- Redshift is based on PostgreSQL, but it's not used for OLTP
- It's OLAP – online analytical processing (analytics and data warehousing)
- 10x better performance than other data warehouses, scale to PBs of data
- **Columnar** storage of data (instead of row based) & parallel query engine
- Two modes: Provisioned cluster or Serverless cluster
- Has a SQL interface for performing the queries
- BI tools such as Amazon Quicksight or Tableau integrate with it
- **vs Athena:** faster queries / joins / aggregations thanks to indexes

Redshift – Snapshots & DR

- Redshift has “Multi-AZ” mode for some clusters
- Snapshots are point-in-time backups of a cluster, stored internally in S3
- Snapshots are incremental (only what has changed is saved)
- You can restore a snapshot into a **new cluster**
- Automated: every 8 hours, every 5 GB, or on a schedule. Set retention between 1 to 35 days
- Manual: snapshot is retained until you delete it
- You can configure Amazon Redshift to automatically copy snapshots (automated or manual) of a cluster to another AWS Region



Amazon OpenSearch Service



- *Amazon OpenSearch is successor to Amazon Elasticsearch*
- In DynamoDB, queries only exist by primary key or indexes...
- **With OpenSearch, you can search any field, even partially matches**
- It's common to use OpenSearch as a complement to another database
- Two modes: managed cluster or serverless cluster
- Does *not* natively support SQL (can be enabled via a plugin)
- Ingestion from Kinesis Data Firehose, AWS IoT, and CloudWatch Logs
- Security through Cognito & IAM, KMS encryption, TLS
- Comes with OpenSearch Dashboards (visualization)

Amazon EMR



- EMR stands for “Elastic MapReduce”
- EMR helps creating **Hadoop clusters (Big Data)** to analyze and process vast amount of data
- The clusters can be made of **hundreds of EC2 instances**
- EMR comes bundled with Apache Spark, HBase, Presto, Flink...
- EMR takes care of all the provisioning and configuration
- Auto-scaling and integrated with Spot instances
- Use cases: data processing, machine learning, web indexing, big data...